

Improving Survival Data Visualization: A Clinician's Perspective

Thank you for participating in this survey.

This survey is conducted by Victor Pacifique Rwandarwacu and Shyamsundar Ranganath as part of the requirements for the MSc in Health Data Science, School of Mathematical and Statistical Sciences at the University of Galway, Ireland.

Survival analysis plays a vital role in medical research; however, the way survival data are visualised can significantly influence clinical decision-making and interpretation. This survey aims to explore how medical doctors perceive various survival data visualisations, including Kaplan-Meier curves, survival difference plots, mean residual life (MRL), and mean residual life difference (MRLD) plots.

You will be presented with different visualisations accompanied by brief interpretations and asked to rank them based on their interpretability. Your insights will contribute to improving how survival data are communicated in both clinical and research contexts.

The survey will take approximately 15 minutes to complete. Participation is voluntary, and all responses will be kept strictly confidential.

Deadline: 7th June 2025

For more information, please contact:

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Shyamsundar Ranganath – s.ranganath1@universityofgalway.ie

Thank you for your time and valuable contribution!

* Indicates required question

Knowledge on survival analysis

Survival analysis is also known as time to event data analysis e.g: time to death, time to recurrence or time to discharge etc.

Rate your knowledge on a scale of 1- 10 where 1 represents no knowledge and 10 represents expert knowledge.

1. Rate your Knowledge about survival analysis on a scale of 1 to 10 *

1	2	3	4	5	6	7	8	9	10
☆	☆	☆	☆	☆	☆	☆	☆	☆	☆

2. Qualification *

Mark only one oval.

- ☐ Medical Doctor (General Practitioner)
- ☐ Medical Doctor (Specialist or postgraduate)
- ☐ Medical student

3. Country *

 Dropdown*Country of undergraduate training**Mark only one oval.*

- ☐ Afghanistan
- ☐ Albania
- ☐ Algeria
- ☐ American Samoa
- ☐ Andorra
- ☐ Angola
- ☐ Anguilla
- ☐ Antarctica
- ☐ Antigua and Barbuda
- ☐ Argentina
- ☐ Armenia
- ☐ Aruba
- ☐ Australia
- ☐ Austria
- ☐ Azerbaijan
- ☐ Bahamas
- ☐ Bahrain
- ☐ Bangladesh
- ☐ Barbados
- ☐ Belarus
- ☐ Belgium
- ☐ Belize
- ☐ Benin
- ☐ Bermuda
- ☐ Bhutan
- ☐ Bolivia, Plurinational State of
- ☐ Bonaire, Sint Eustatius and Saba
- ☐ Bosnia and Herzegovina
- ☐ Botswana
- ☐ Bouvet Island
- ☐ Brazil
- ☐ British Indian Ocean Territory

- ☐ Brunei Darussalam
- ☐ Bulgaria
- ☐ Burkina Faso
- ☐ Burundi
- ☐ Cabo Verde
- ☐ Cambodia
- ☐ Cameroon
- ☐ Canada
- ☐ Cayman Islands
- ☐ Central African Republic
- ☐ Chad
- ☐ Chile
- ☐ China
- ☐ Christmas Island
- ☐ Cocos (Keeling) Islands
- ☐ Colombia
- ☐ Comoros
- ☐ Congo
- ☐ Congo, The Democratic Republic of the
- ☐ Cook Islands
- ☐ Costa Rica
- ☐ Croatia
- ☐ Cuba
- ☐ Curaçao
- ☐ Cyprus
- ☐ Czechia
- ☐ Côte d'Ivoire
- ☐ Denmark
- ☐ Djibouti
- ☐ Dominica
- ☐ Dominican Republic
- ☐ Ecuador
- ☐ Egypt
- ☐ El Salvador
- ☐ Equatorial Guinea

- ☐ Eritrea
- ☐ Estonia
- ☐ Eswatini
- ☐ Ethiopia
- ☐ Falkland Islands (Malvinas)
- ☐ Faroe Islands
- ☐ Fiji
- ☐ Finland
- ☐ France
- ☐ French Guiana
- ☐ French Polynesia
- ☐ French Southern Territories
- ☐ Gabon
- ☐ Gambia
- ☐ Georgia
- ☐ Germany
- ☐ Ghana
- ☐ Gibraltar
- ☐ Greece
- ☐ Greenland
- ☐ Grenada
- ☐ Guadeloupe
- ☐ Guam
- ☐ Guatemala
- ☐ Guernsey
- ☐ Guinea
- ☐ Guinea-Bissau
- ☐ Guyana
- ☐ Haiti
- ☐ Heard Island and McDonald Islands
- ☐ Holy See (Vatican City State)
- ☐ Honduras
- ☐ Hong Kong
- ☐ Hungary
- ☐ Iceland

- ☐ India
- ☐ Indonesia
- ☐ Iran, Islamic Republic of
- ☐ Iraq
- ☐ Ireland
- ☐ Isle of Man
- ☐ Israel
- ☐ Italy
- ☐ Jamaica
- ☐ Japan
- ☐ Jersey
- ☐ Jordan
- ☐ Kazakhstan
- ☐ Kenya
- ☐ Kiribati
- ☐ Korea, Democratic People's Republic of
- ☐ Korea, Republic of
- ☐ Kuwait
- ☐ Kyrgyzstan
- ☐ Lao People's Democratic Republic
- ☐ Latvia
- ☐ Lebanon
- ☐ Lesotho
- ☐ Liberia
- ☐ Libya
- ☐ Liechtenstein
- ☐ Lithuania
- ☐ Luxembourg
- ☐ Macao
- ☐ Madagascar
- ☐ Malawi
- ☐ Malaysia
- ☐ Maldives
- ☐ Mali
- ☐ Malta

- ☐ Marshall Islands
- ☐ Martinique
- ☐ Mauritania
- ☐ Mauritius
- ☐ Mayotte
- ☐ Mexico
- ☐ Micronesia, Federated States of
- ☐ Moldova, Republic of
- ☐ Monaco
- ☐ Mongolia
- ☐ Montenegro
- ☐ Montserrat
- ☐ Morocco
- ☐ Mozambique
- ☐ Myanmar
- ☐ Namibia
- ☐ Nauru
- ☐ Nepal
- ☐ Netherlands
- ☐ New Caledonia
- ☐ New Zealand
- ☐ Nicaragua
- ☐ Niger
- ☐ Nigeria
- ☐ Niue
- ☐ Norfolk Island
- ☐ North Macedonia
- ☐ Northern Mariana Islands
- ☐ Norway
- ☐ Oman
- ☐ Pakistan
- ☐ Palau
- ☐ Palestine, State of
- ☐ Panama
- ☐ Papua New Guinea

- ☐ Paraguay
- ☐ Peru
- ☐ Philippines
- ☐ Pitcairn
- ☐ Poland
- ☐ Portugal
- ☐ Puerto Rico
- ☐ Qatar
- ☐ Romania
- ☐ Russian Federation
- ☐ Rwanda
- ☐ Réunion
- ☐ Saint Barthélemy
- ☐ Saint Helena, Ascension and Tristan da Cunha
- ☐ Saint Kitts and Nevis
- ☐ Saint Lucia
- ☐ Saint Martin (French part)
- ☐ Saint Pierre and Miquelon
- ☐ Saint Vincent and the Grenadines
- ☐ Samoa
- ☐ San Marino
- ☐ Sao Tome and Principe
- ☐ Saudi Arabia
- ☐ Senegal
- ☐ Serbia
- ☐ Seychelles
- ☐ Sierra Leone
- ☐ Singapore
- ☐ Sint Maarten (Dutch part)
- ☐ Slovakia
- ☐ Slovenia
- ☐ Solomon Islands
- ☐ Somalia
- ☐ South Africa
- ☐ South Georgia and the South Sandwich Islands

- ☐ South Sudan
- ☐ Spain
- ☐ Sri Lanka
- ☐ Sudan
- ☐ Suriname
- ☐ Svalbard and Jan Mayen
- ☐ Sweden
- ☐ Switzerland
- ☐ Syrian Arab Republic
- ☐ Taiwan, Province of China
- ☐ Tajikistan
- ☐ Tanzania, United Republic of
- ☐ Thailand
- ☐ Timor-Leste
- ☐ Togo
- ☐ Tokelau
- ☐ Tonga
- ☐ Trinidad and Tobago
- ☐ Tunisia
- ☐ Turkey
- ☐ Turkmenistan
- ☐ Turks and Caicos Islands
- ☐ Tuvalu
- ☐ Uganda
- ☐ Ukraine
- ☐ United Arab Emirates
- ☐ United Kingdom
- ☐ United States
- ☐ United States Minor Outlying Islands
- ☐ Uruguay
- ☐ Uzbekistan
- ☐ Vanuatu
- ☐ Venezuela, Bolivarian Republic of
- ☐ Viet Nam
- ☐ Virgin Islands, British

- ☐ Virgin Islands, U.S.
- ☐ Wallis and Futuna
- ☐ Western Sahara
- ☐ Yemen
- ☐ Zambia
- ☐ Zimbabwe
- ☐ Åland Islands

PART 1

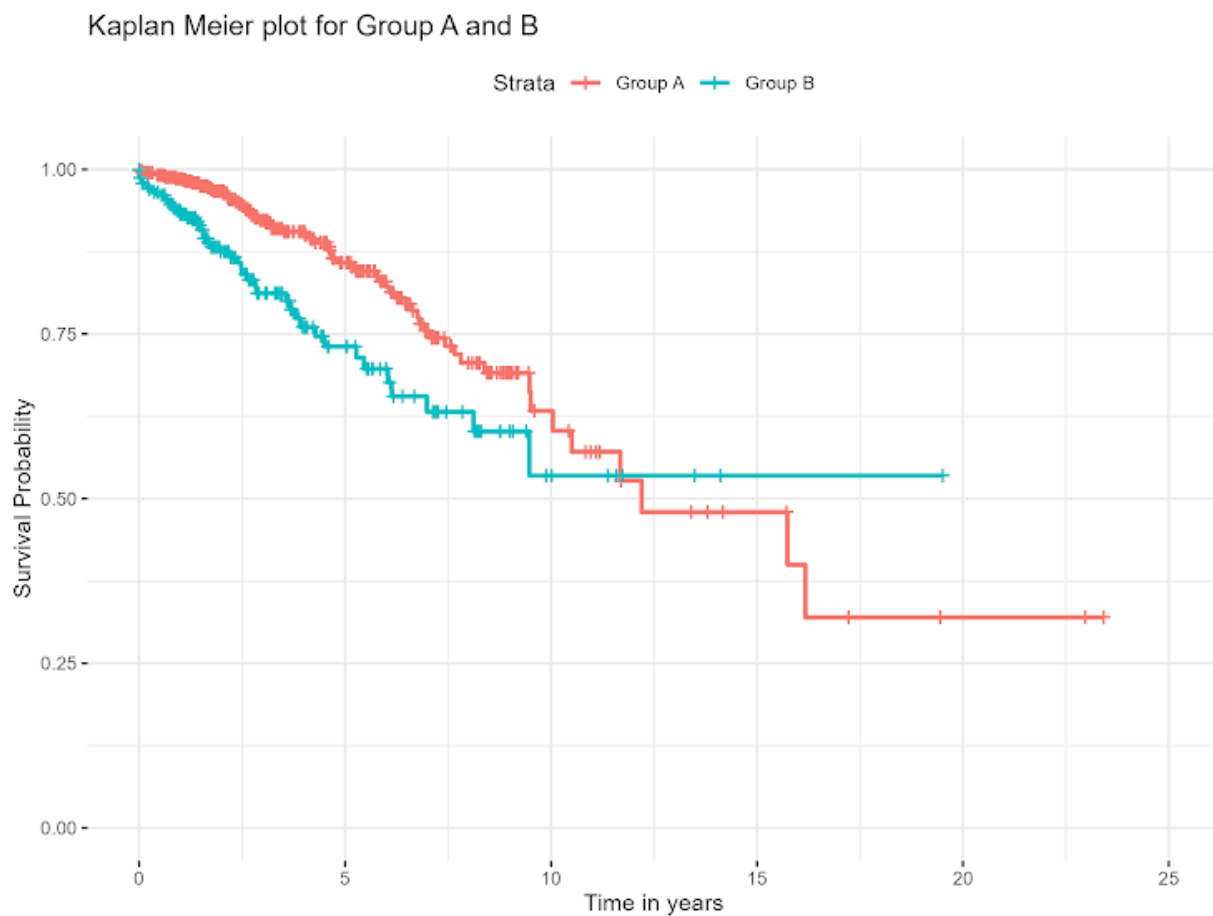
In this section, you will be presented with various visualisations of time-to-event (survival) data.

You will be asked to review each plot and provide your interpretation or insights. Please write down any observations or thoughts you have regarding the data presentation, trends, or patterns you notice.

4. What is your interpretation of this Kaplan Meier (KM) Survival Plot? *

Please describe any trends, differences between groups, or key observations you notice in the plot.

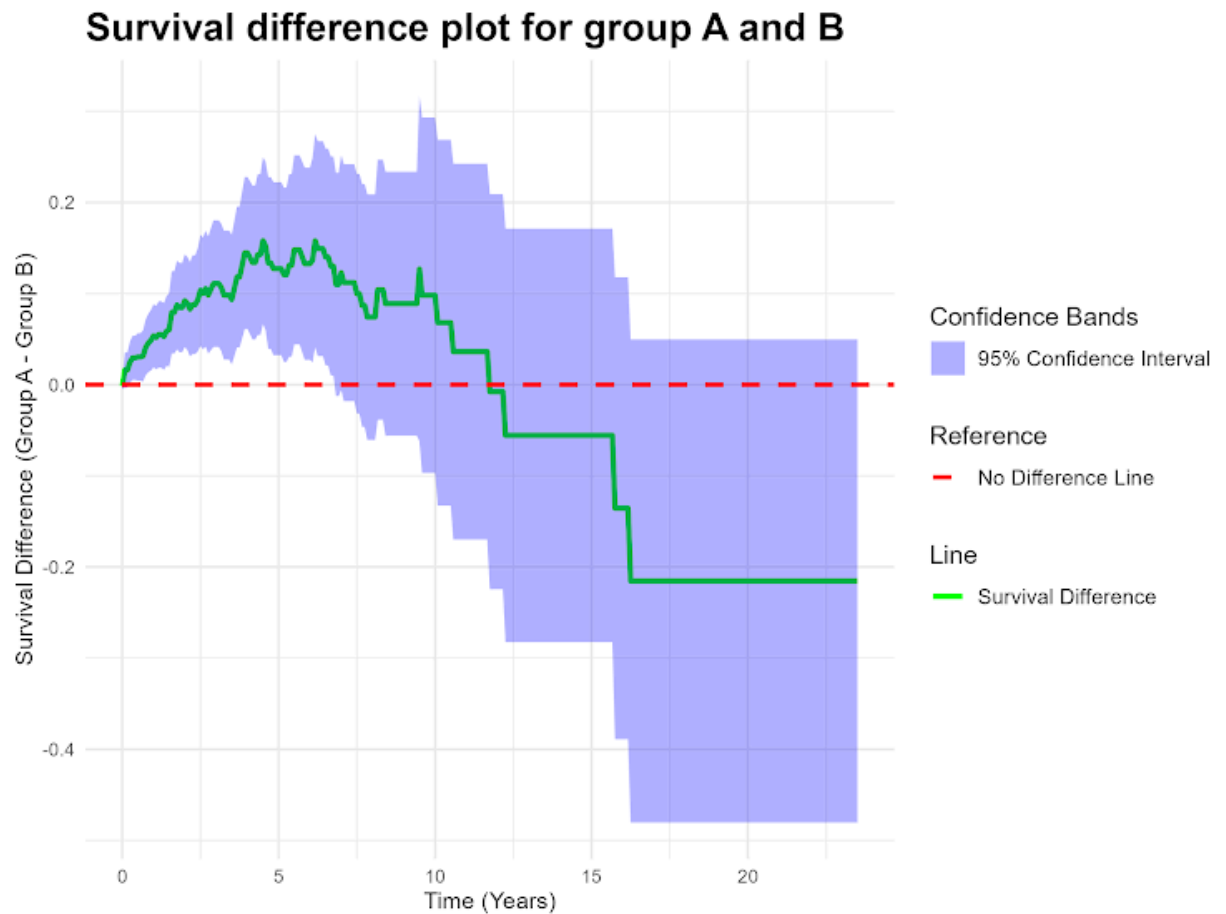
Note: Interpretations will be provided in the next section.



5. **What is your interpretation of this plot of the difference in survival between the two groups?** *

Please share any insights, trends, or differences you observe between the groups over time.

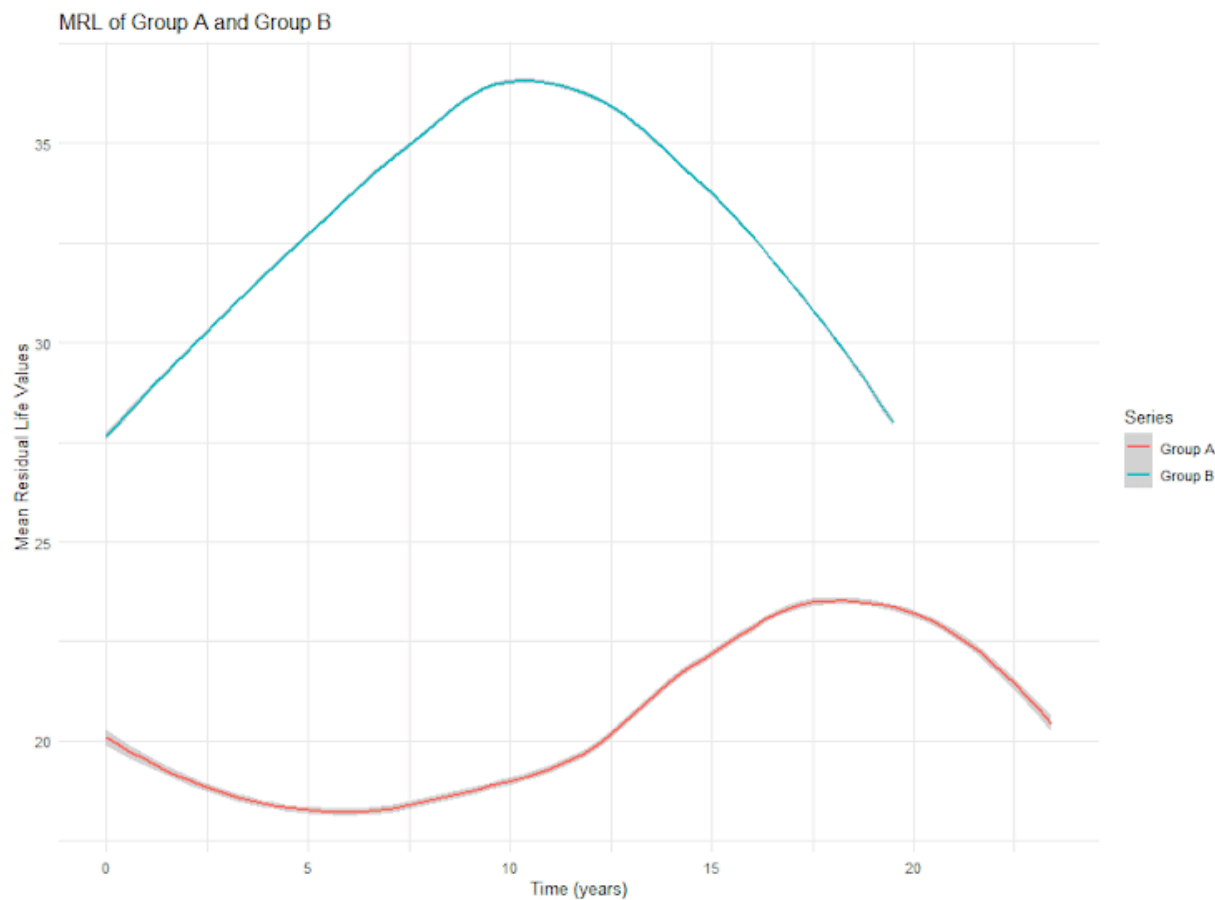
Note: Interpretations will be provided.



6. What is your interpretation of this Mean Residual Life (MRL) plot? *

Please describe any patterns, group differences, or key insights you observe regarding expected remaining survival time.

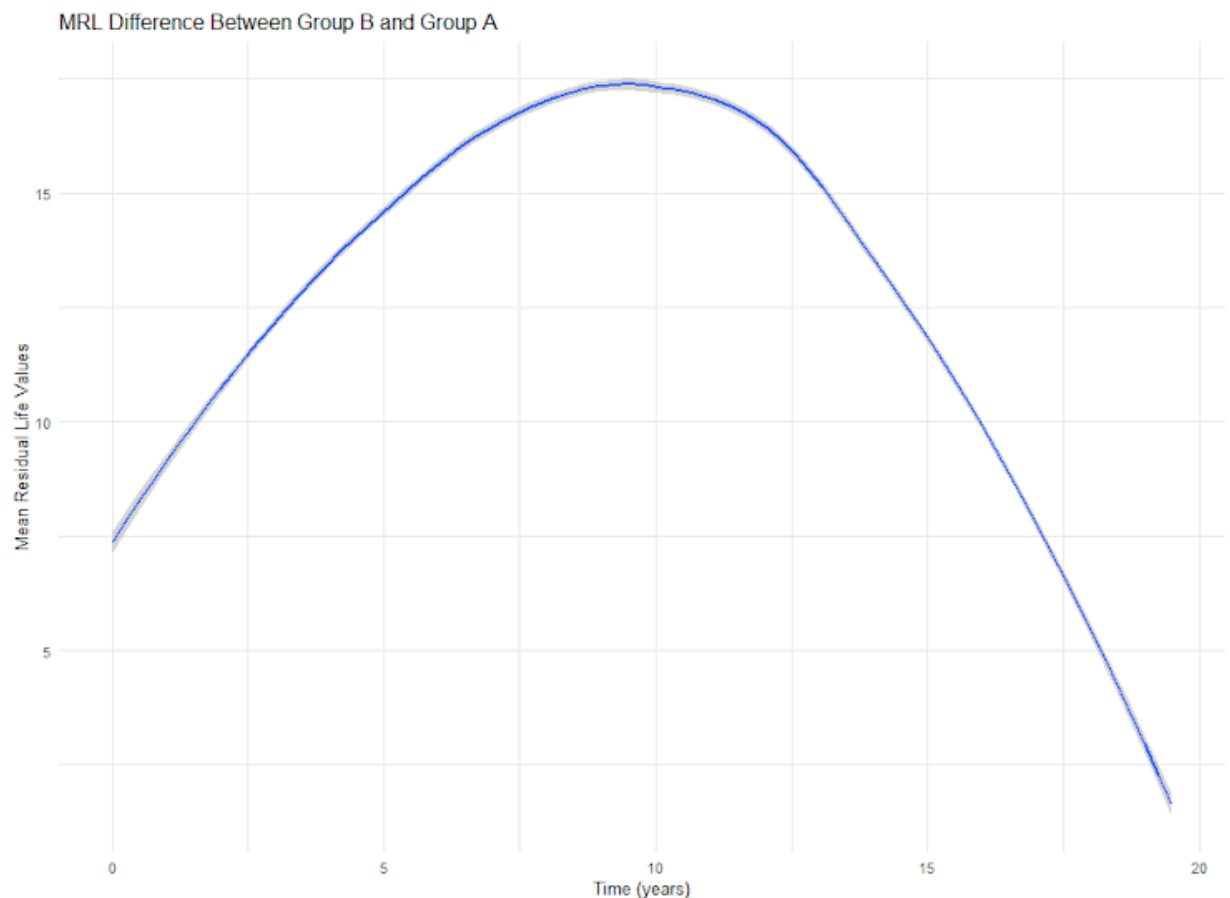
Note: Interpretations will be provided.



7. **What is your interpretation of this plot of the difference in Mean Residual Life between the groups?** *

Please comment on any notable differences between groups, including changes in the expected remaining survival time over the follow-up period.

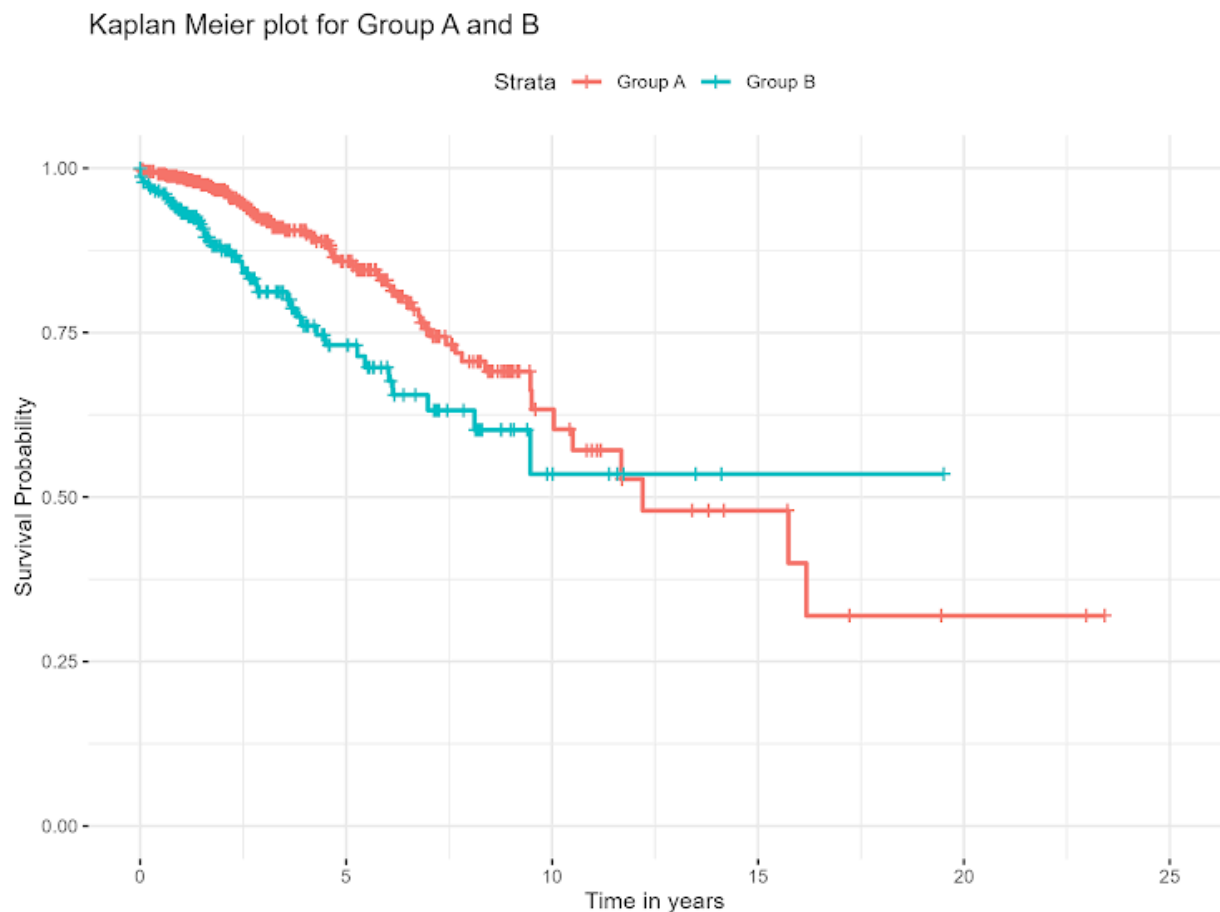
Note: Interpretations will be provided.



In this section, we provide interpretations of the survival plots presented earlier.

These explanations offer insights into how each plot can be understood and highlight key differences, trends, or clinical implications that may be observed.

1. Interpretation : Kaplan Meier Plot



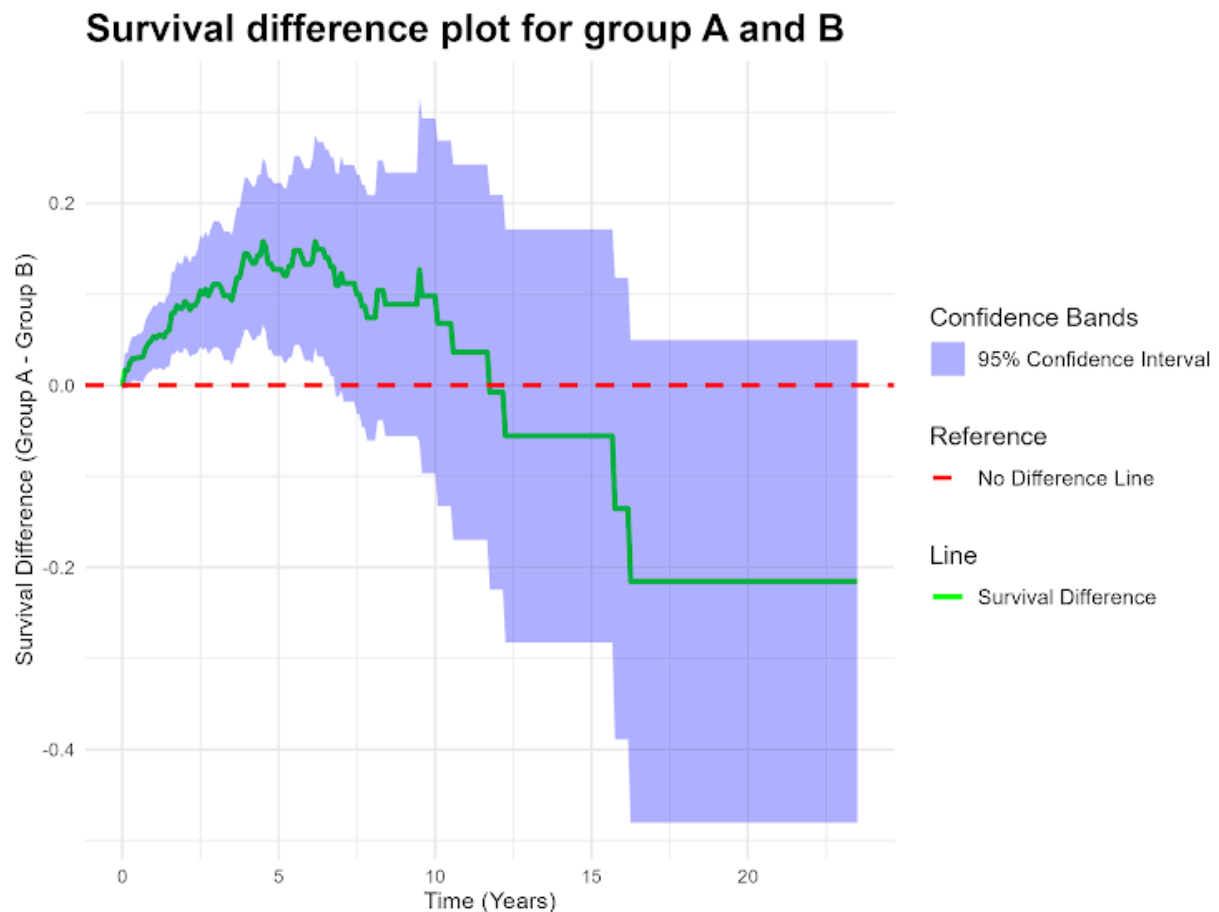
Interpretation : Kaplan Meier Plot

This KM curve compares survival between two patients' groups (Group A and Group B) over time.

From the start, Patients in Group A had better survival compared to Group B suggesting a better long-term outcome. For example, 5 years post-diagnosis 73% of Group B patients were still alive compared to approximately 84% of Group A patients.

Clinically, this suggests a survival advantage for Group A over Group B.

Interpretation: Survival Difference Plot



Interpretation : Survival difference Plot

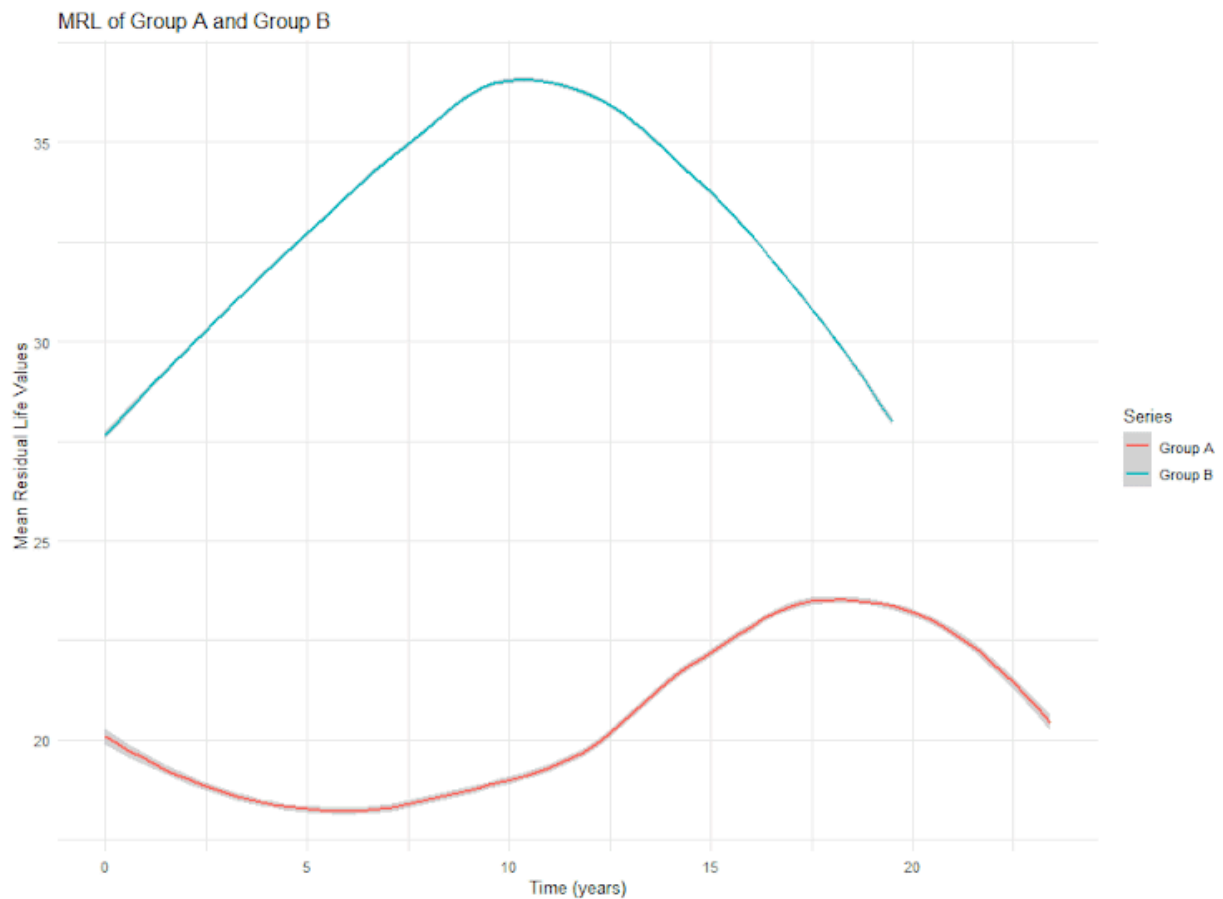
This plot shows the difference in survival between Group A and Group B (Group A - Group B) over time, displayed as a green line, which highlights the improved survival prospects for Group A.

In the first 7 years post diagnosis, Group A has better survival (as seen by the green line and band staying above zero), but this advantage declines later.

The shaded area (in blue) represents the 95% Confidence Interval (CI) for the difference in survival. If the CI does not contain zero there is evidence that the difference in survival is likely to be real and not due to chance. In this example there is evidence of improved survival for Group A from 0 to 7 years approximately as the 95% CI is strictly positive across this range.

Clinically, this suggests that while Group A initially had a more favorable prognosis compared to Group B, the long-term benefit may not be sustained.

Interpretation : Mean Residual Life (MRL) Plot



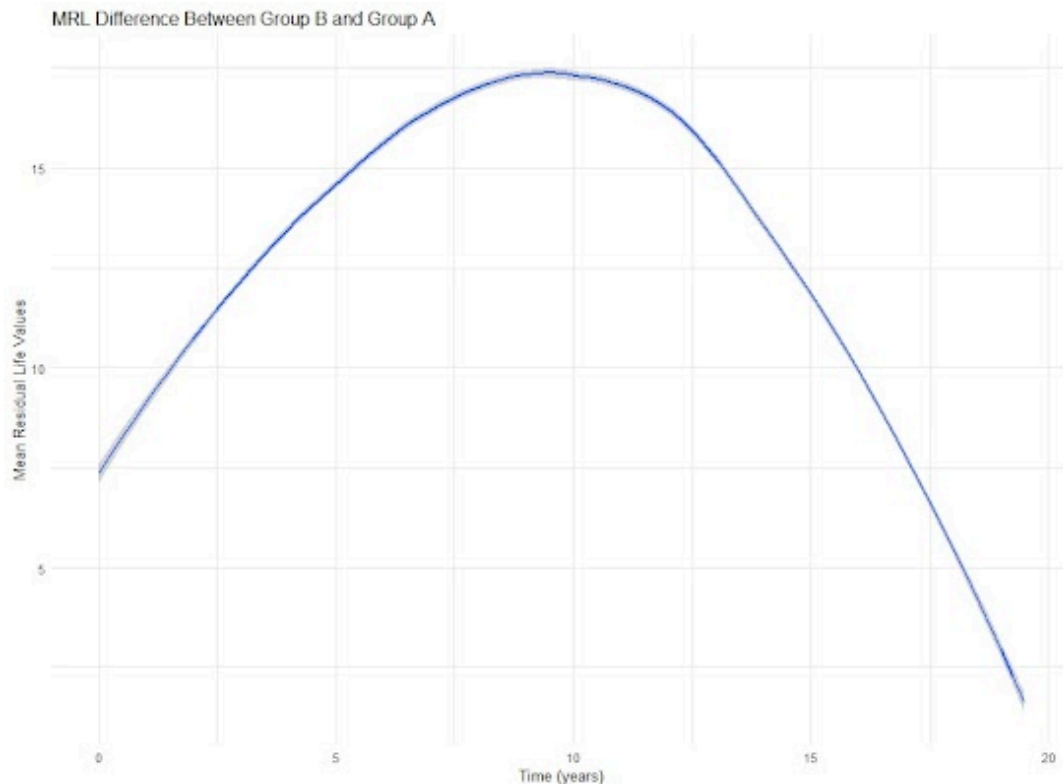
Interpretation : Mean Residual Life Plot

The MRL plot displays the estimated average remaining lifetime for patients who have already survived up to a specific time e.g. on average how much time have I left given I have survived 2 years?

For example, given that patient is still alive 10 years post diagnosis , the expected remaining life for Group B is about 36 years, compared to only about 19 years for Group A.

Clinically, this graph highlights that patients in Group B have a higher mean residual life compared to Group A.

Interpretation : Mean Residual Life Difference Plot



Interpretation : Mean Residual Life Difference Plot

The MRL difference plot quantifies how much longer patients in Group B are expected to live compared to those in group A given that both have survived up to a certain time point post diagnosis.

Group B consistently has a better prognosis, with the difference peaking around year 10 at nearly 17 years. This means 10 years post diagnosis, individuals in Group B can expect to live, on average, 17 years longer than those in Group A. However, the groups become more similar in their remaining life expectancy in later years.

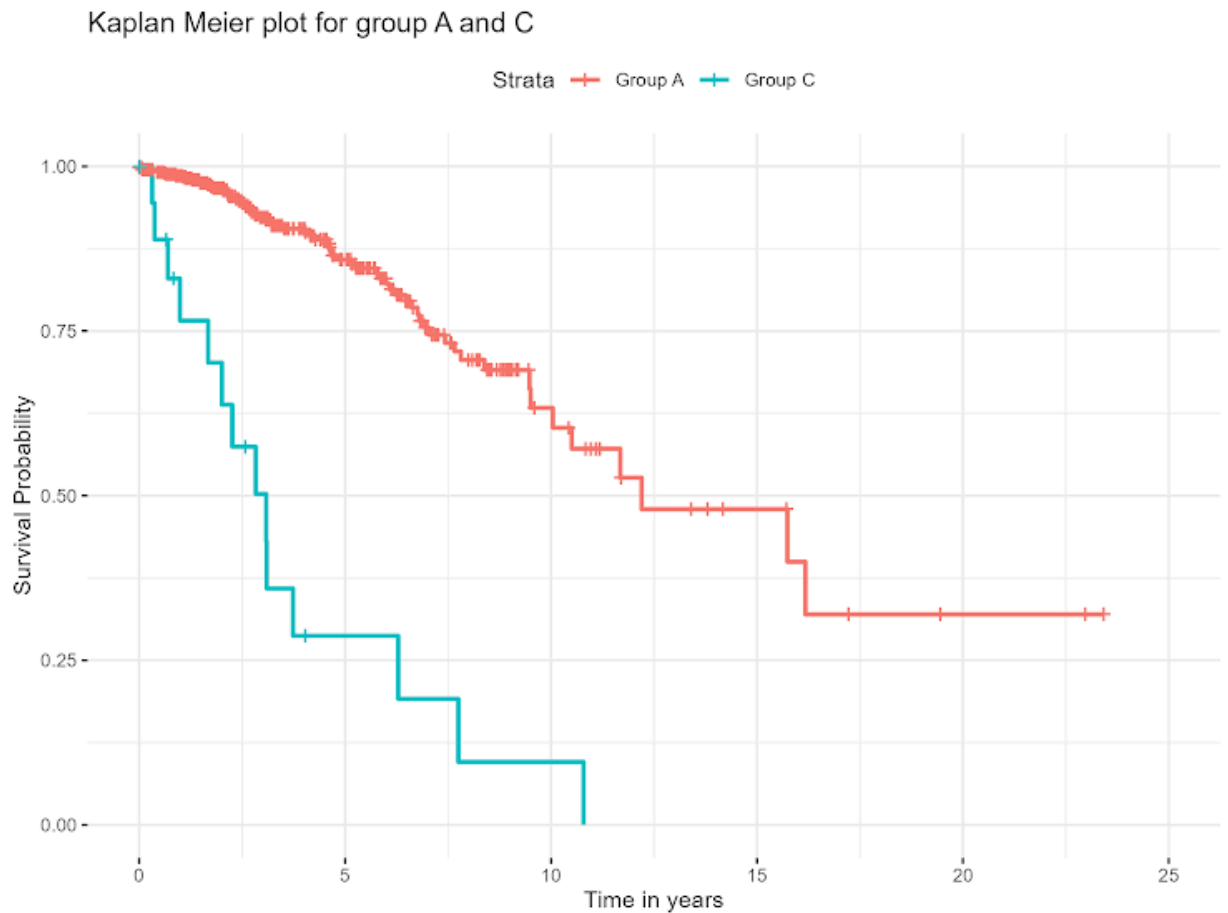
Apply Your Insight

You've seen example plots with their respective interpretations. Now, we invite you to review a new set of plots and share your own insights.

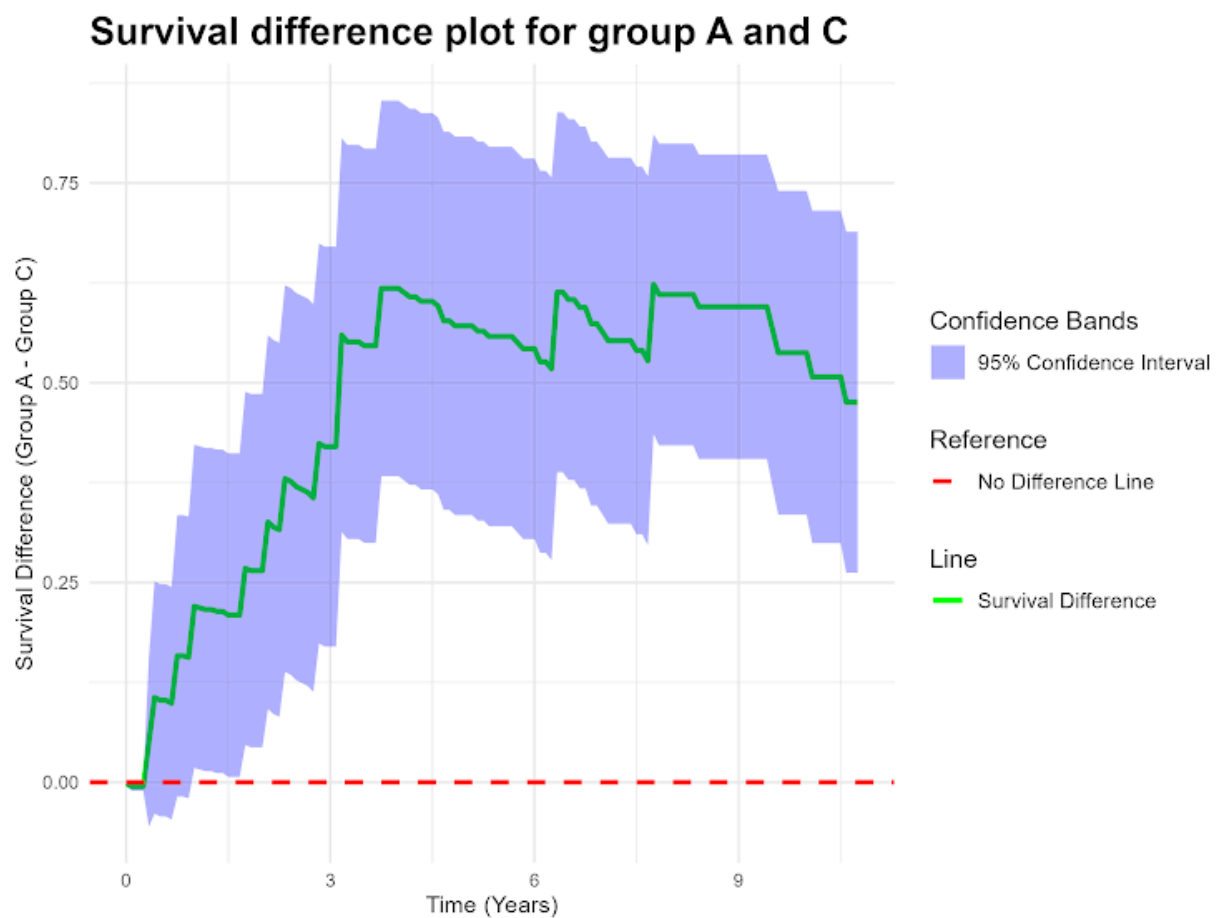
Please describe any patterns, differences, or key observations you notice in each visualisation.

8. what is your insight or interpretation of this image? *

Feel free to write any comment, In less than five sentences

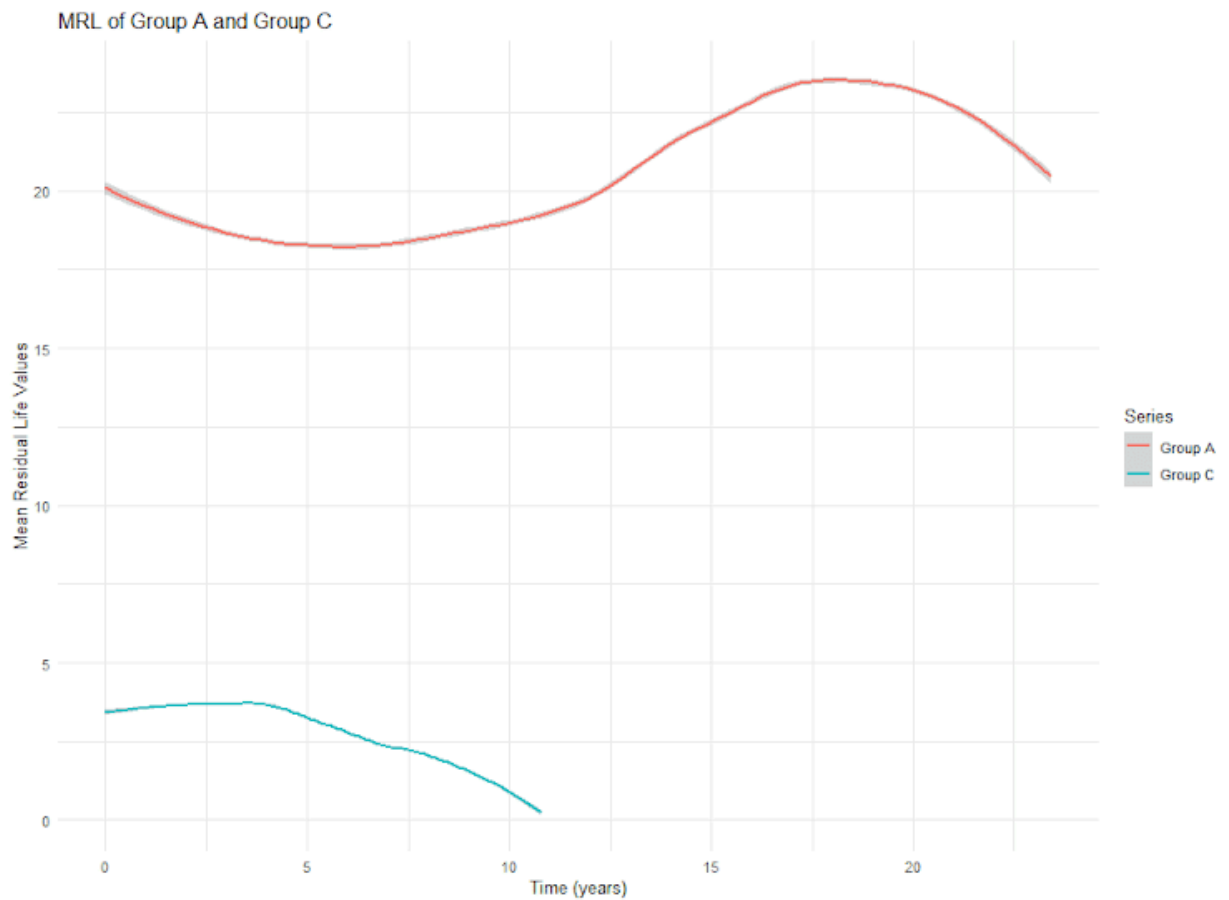


9. what is your insight or interpretation of this image? *



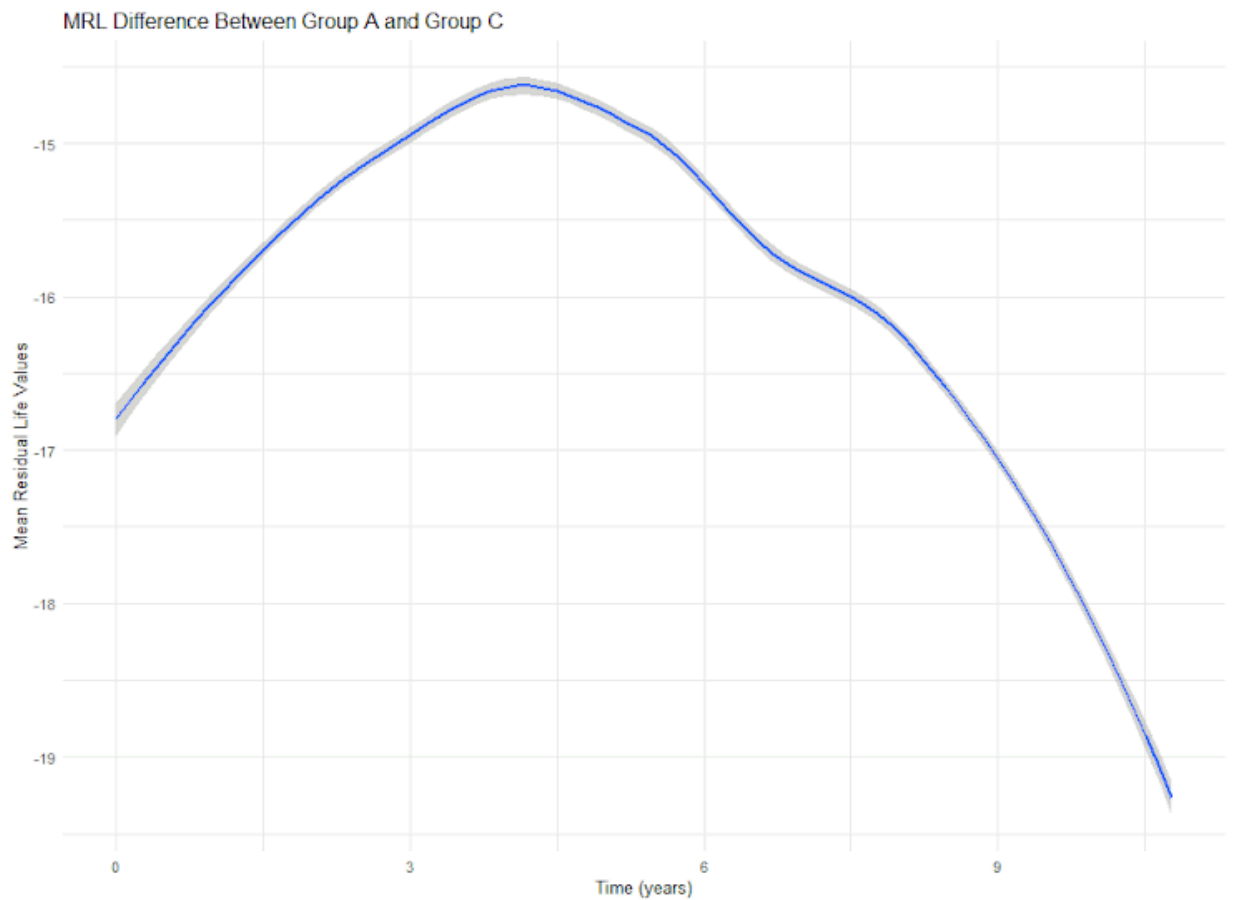
10. **MEAN RESIDUAL LIFE PLOT**

*

what is your insight or interpretation of this image?***(In less than five sentences)***

11. **MEAN RESIDUAL LIFE DIFFERENCE**

*

what is your insight or interpretation of this image?***(In less than five sentences)*****RANKING OF THE PLOTS***Rank the plots by your preference*

12. Which plot did you find easy to understand ? *

Rank from the easiest (1) to the hardest (4)

Mark only one oval per row.

	1	2	3	4
Kaplan Meier plot	☐	☐	☐	☐
Survival Difference plot	☐	☐	☐	☐
Mean Residual Life plot	☐	☐	☐	☐
Mean Residual Life Difference plot	☐	☐	☐	☐

13. Which plot would you prefer to use in consultation with a patients? *

Rank them from the first choice (1) to the last choice (4)

Mark only one oval per row.

	1	2	3	4
Kaplan Meier plot	☐	☐	☐	☐
Survival Difference plot	☐	☐	☐	☐
Mean Residual Life plot	☐	☐	☐	☐
Mean Residual Life Difference plot	☐	☐	☐	☐

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